

POSTULAE - Backend Development Handoff Note

Confidential - v1.0 - February 2026 - For Development Agency Only

1. Project Overview

Postulae is a SaaS platform that **scores and generates elite one-page CVs** for French *grande ecole* students targeting finance, consulting, and selective startup roles. Two core engines: a **freemium CV grader** (score /100, captures emails and drives sign-ups) and a **paid CV generator** (optimized PDF + DOCX in FR and EN). The frontend is nearly complete. This note covers everything needed to build the backend (FastAPI, DB, auth, Stripe, algorithms).

2. Shared Drive Contents

Path	Description
cv_enhancer_v0226/app/	Production Python modules: generator, grader, LLM client, layout, density, enrichment
app/prompts/	LLM prompt files: base_system.txt (28KB detailed prompt), extract_from_pdf.txt, enrich_content.txt
app/templates/grid_template.html	Current HTML/CSS CV layout, serves as a starting point (see layout notes in section 5)
CLAUDE.md	Full technical history: architecture, resolved bugs, PFR calibration, session logs
README.md	Usage guide, API examples, module responsibilities, dependency list
Perfect_CV_FR/ENG .pdf/.docx	Reference "gold standard" CVs, the target output quality (see benchmarks in section 9)
BAD_CV.png	Example of a poor CV (photo, colors, 2-column layout), must score low in grader

3. Codebase Architecture (app/ folder)

Module	Role
generator.py	Main orchestrator: extract > generate > measure PFR > enrich/trim > export PDF+DOCX
llm_client.py	All OpenAI calls (GPT-4o). Extraction via Vision API, content generation, fallback logic
cv_grader.py	Freemium scoring /100: categories, hard rules (caps), tips + CTA. Deterministic, no LLM
content_analyzer.py	Classifies source CV richness (rich/medium/poor/critical), sets enrichment strategy
density.py	PFR calculation from rendered PDF via pdfplumber. Thresholds adjustable (see section 5)
enrichment.py	Single-pass content expansion or trimming. Current approach avoids loops to control costs
layout.py	Renders CV data into HTML template > PDF (xhtml2pdf) > DOCX (pdf2docx). Jinja2 templating
models.py	Pydantic v2 models: CVContent, CVGenerationResult, PageFillMetrics

4. Plans, Pricing and User Roles

Plan	Price	Backend Capabilities
Essentiel	Free	Upload CV > score /100 + 3 tips + CTA. Email captured before upload. No PDF generated.
Starter	4.90/mo	Single CV generation: PDF + DOCX in FR and EN. Stripe checkout before generation.
Premium	24.90/mo	Unlimited CV + cover letter generation. Editable DOCX. Document history. Full dashboard.
Ultimate	149 (one-time)	Tutor review by MBB/Big 4 profile. Rewritten CV + cover letter, commented. Priority support.
Tutor (internal)	Invited	Reviews Ultimate orders. Sees assigned CVs, provides manual feedback, validates output.
Admin (internal)	Full access	User/tutor management, analytics, payment monitoring, system config.

5. Hard Rules vs. Adjustable Guidelines

NON-NEGOTIABLE (2 absolute rules):

- 1. Output must always be exactly ONE page.** This is the core product promise. A Postulae CV is always one A4 page, no exceptions. The entire density system (PFR, enrichment, trimming) exists to enforce this.
- 2. Stripe payment must complete BEFORE any generation call.** Never trigger LLM calls before payment confirmation. This protects against cost leaks on failed or cancelled payments.

GUIDELINES - Current approach, open to discussion with dev:

- 1. LLM call limits:** The current design caps at 1 extraction + 1 generation + 1 enrichment per language (4 to 6 calls total for FR+EN). This was implemented to avoid runaway costs from a previous loop-based version. The principle of cost control matters, but the exact limits can be discussed if there's a better approach.
- 2. Page Fill Rate (PFR) thresholds:** Currently set at 86-95% optimal zone, block below 40%. These values were calibrated on the current template but **may need adjustment** depending on your implementation. The real goal is a full, dense, professional-looking A4 page. The exact percentage is a means to that end.
- 3. HTML/CSS template and column layout:** The current grid_template.html uses fixed columns (12% date / 70% content / 18% location). **Desired improvement:** bullet points should extend across the full page width, flowing under the location column (e.g. under "Paris, France") rather than being confined to the 70% content column. The reference CVs (Perfect_CV_FR/ENG) show the target result. The dev is free to rework the CSS/HTML to achieve this.
- 4. LLM model choice:** Current code uses GPT-4o for both extraction and generation. This can be discussed: GPT-4o-mini may be sufficient for extraction or for the freemium grader. Evaluate cost vs. quality tradeoffs.
- 5. Grader scoring categories:** The current breakdown (Structure 25pts, Experience 35pts, Education 15pts, Skills 15pts, Contact 10pts) is a starting point. The dev can propose a different distribution. **What IS fixed:** the hard rules. A CV with photo/colors must score very low (20-35), a 2-page CV must be penalized, and a Postulae-generated CV must score 90+. These ensure the grader drives conversion.
- 6. Language and prompts:** The prompt files (base_system.txt is 28KB) encode the current CV generation style. The dev can refine wording, adjust instructions, or restructure prompts, as long as output quality matches the reference CVs provided in the Drive.

6. Generation Pipeline Overview

Step	Action	Module	LLM?
1	User uploads PDF (after Stripe payment)	FastAPI endpoint	No
2	Extract text from PDF	llm_client.py (GPT-4o Vision)	1 call
3	Analyze source richness, set strategy	content_analyzer.py	No
4	Generate structured CV content (FR)	llm_client.py	1 call
5	Render HTML, measure page fill	layout.py + density.py	No
6	If underfilled: enrich. If overfilled: trim.	enrichment.py	0-1
7	Export PDF + DOCX (FR)	layout.py	No
8	Repeat steps 4-7 for EN (independent PFR)	Same modules	1-2
9	Return results + warning level to frontend	FastAPI response	No

FR and EN are generated independently because French text is about 10-15% longer than English for the same content. Each language may need different enrichment/trimming. Current estimated cost: around \$0.08-0.15 per CV.

7. User-Facing Warning System

When the source CV lacks detail, the engine adds content to fill the page. The API returns a warning level so the frontend can alert the user. Current levels (thresholds adjustable):

Level	Source Quality	Invention	User Message
success (green)	Rich (2500+ chars)	< 10%	"Light optimizations applied"
warning (orange)	Medium (1800-2500)	10-30%	"Significant enrichments, review before use"
error (red)	Poor (1200-1800)	30-50%	"Substantial content inferred, PERSONALIZE"
critical (dark red)	Critical (600-1200)	50-70%	"MASSIVELY inferred, DO NOT send as-is"
block	Empty (< 600 chars)	N/A	"Provide a more detailed CV" (no output generated)

8. Freemium Grader: Key Requirements

Must be idempotent: same CV uploaded twice = exact same score. **Must be cost-efficient:** this endpoint is hit by every visitor. Evaluate whether pdfplumber extraction alone works or if a cheap LLM call (GPT-4o-mini, temperature=0) is needed. **Hard rules are fixed:** CV with photo/colors = score 20-35. 2-page CV = heavily penalized. Postulae-generated CV = must score 90+. **Known bug:** reference CV currently scores around 80 instead of 95+ (field mapping: bullets vs responsibilities), fix during integration. Category weights and tip wording are open to refinement.

9. Reference Benchmarks and Tech Stack

Reference CV	Grader Target	Generator Target	Notes
Perfect_CV_FR.pdf	95+	Full dense A4 page	Gold standard. 4 experiences, 14 bullets. Named "Fafa SELECTO".
Perfect_CV_ENG.pdf	95+	Full dense A4 page	English version. Slightly shorter text, may need adjustment.
BAD_CV.png	20-35	N/A	Photo, colors, 2-column layout. Grader must penalize heavily.

Stack: FastAPI, Python 3.12+, OpenAI GPT-4o, pdfplumber, xhtml2pdf, pdf2docx, Jinja2, Pydantic v2, Stripe. Engine is stateless: add your own storage layer for generated files.